MESHDECK

CR // OS 2.0

An offline field operating system for the M5Stack Tab5

User manual: what it does, how to use it, and what it can and cannot do

Version 2.0.0-dev (V7) with GAMES (RALLY, SEA BATTLE, KLONDIKE), TOOLS, the EMCON radio levels and the boot splash.

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1. ABOUT MESHDECK

MESHDECK is an offline field operating system. It runs on an M5Stack Tab5, a 5-inch touchscreen computer with a physical keyboard, and it is built for civilian resilience: staying in touch, finding your way, keeping records and passing the time when there is no internet and no phone network.

MESHDECK does five jobs:

- Talk. Messages over a Meshtastic mesh radio: chat with a channel or a single node.
- Find. Offline Ordnance Survey maps, a search of about 25,000 places, and the positions of other nodes.
- Read and keep. A reference library, a book reader for PDF and EPUB books, a field log, reports, an inventory and notes.
- Run. A terminal of typed commands, a diagnostics page and settings.
- Lock. A PIN lock with an auto-lock time, so the deck can be left without being used by someone else.

There is also a small set of games for downtime (section 14). Everything works with no internet and no phone network. The deck does not connect to the internet, and nothing is sent anywhere unless you send it yourself over the mesh (section 18).

! Not yet built. Some features are not in this version. Section 20 lists them in full.

2. THE HARDWARE YOU NEED

PART	NEEDED?	WHAT IT IS FOR
M5Stack Tab5	Yes	The computer: 5-inch 1280 x 720 touchscreen, physical keyboard, battery, USB-C.
microSD card (FAT32)	Yes	Everything you save: messages, maps, books, files, logs, reports, notes. Without a card the deck still runs, but nothing is kept.
M5Stack Unit C6L (Meshtastic radio)	For messages and positions	Plugs into the USB-A port. Connects MESHDECK to a Meshtastic mesh network.
Map packs and place list	For maps and place search	Copied to the card (section 7). Not supplied in the download.
Book packs	For the book reader	Copied to the card, or made on the deck from an EPUB (section 9).

> MESHDECK is developed for the Tab5. The keyboard must be a physical one: there is no on-screen keyboard. Touch works for buttons, tiles and lists; typing needs the physical keyboard.

3. YOUR FIRST START

The first time MESHDECK starts, the setup walks you through three steps. Each step can be skipped and changed later.

1. Welcome. A short introduction.
2. Clock. Set the date and time. MESHDECK uses UK time, and works out GMT or BST for you. The clock keeps time on its own battery, so you set it once.
3. PIN. Choose whether to protect the deck with a PIN (4 to 12 digits). You can set one later in SECURITY.

After setup you reach the home screen (section 4). The splash (section 15) shows at every power-on before that.

4. THE HOME SCREEN AND THE KEYS

The home screen has eleven tiles, each opened by tapping it or pressing its key:

KEY	TILE	WHAT IT OPENS
1	MESSAGES	Mesh chat, nodes and contacts (section 6)
2	MAP	Offline maps, place search, positions (section 7)
3	LIBRARY	Reference cards and the book reader (section 8)
4	LOGBOOK	Field log, reports, inventory and notes (section 9)
5	FILES	Your files, screenshots and imports (section 10)
6	TERMINAL	Typed commands (section 11)
7	SETTINGS	Display, clock and about (section 12)
8	DIAGNOSTICS	Power and system figures (section 13)
9	SECURITY	PIN, auto-lock and lock now (section 5)
0	GAMES	RALLY, SEA BATTLE and KLONDIKE (section 14)
A	TOOLS	Timer, stopwatch and calculator (section 14a)

The empty space beside the tiles shows the radio and the deck (section 15). The last space shows the time, the date and the battery time left.

4.1 KEYS

KEY	DOES
1 to 9, 0, A	Open a home tile from anywhere. On a screen with its own numbered parts (LOGBOOK, SETTINGS, GAMES), the numbers open those parts first.
Arrows	Move between buttons and places. In games and the map, they move the cursor or the picture.
Enter	Press the highlighted button or open the highlighted item.
Esc	Back one step. From a tile it returns home.
+ and -	Zoom the map or turn the page, where the screen uses them. Brightness is set in SETTINGS.
Ctrl+P	Save a screenshot to MESHDECK/Files/Screenshots (it appears at the top of FILES).
Ctrl+N	Change the colour theme: green, amber or red.
Ctrl+L	Lock now (needs a PIN set).

The line at the bottom of every screen says which keys do what on that screen.

4.2 TOUCH

Tap a tile, a button, a list item or a place to use it. Scroll lists by dragging. The Back button at the top left returns one step. Where a screen needs typed text, use the physical keyboard.

5. SECURITY: THE PIN AND THE LOCK

SECURITY (tile 9) sets, changes and removes the PIN, sets the auto-lock time, and sets the radio emissions (EMCON).

- PIN: 4 to 12 digits. The PIN itself is never stored; MESHDECK keeps a salted PBKDF2-HMAC-SHA256 hash of it (20,000 rounds), in the Tab5's internal flash, not on the card. Taking the card out does not remove the PIN.
- When it locks: at start-up when a PIN is set; after the auto-lock time with no key or touch (OFF, 2, 5 or 15 minutes; 5 by default); and on Ctrl+L or LOCK NOW.
- Wrong PINs: the first three in a row are free. Each further wrong PIN makes you wait: 30 seconds, then 1 minute, 5 minutes, 15 minutes, then an hour each time. The count is saved, so a restart does not skip the wait. Nothing is ever wiped automatically.
- Radio emissions (EMCON): NORMAL sends and receives. RX ONLY sends nothing: messages wait as HELD, and the radio's own transmitter is switched off, so it does not relay either. RADIO OFF stops MESHDECK using the radio and switches its transmitter off; the radio stays powered, because MESHDECK never powers hardware down. The choice is kept across restarts. The Security screen shows the current level and what it means.
- While locked: only the lock screen responds. Over USB serial only STATUS, POWER and KEY (for typing the PIN) are accepted.

! Limits. The files on the SD card are not encrypted. Anyone who takes the card out can read it on a PC. The lock stops someone using the deck, not someone with a card reader. A four-digit PIN can be guessed over days with the waits above, so use six or more digits if that matters.

6. MESSAGES: MESH CHAT

MESSAGES talks to other Meshtastic nodes through the Unit C6L radio. It needs the radio plugged into the USB-A port; without it the screen says NO RADIO and explains what to connect.

6.1 THE SCREENS

SCREEN	WHAT IT SHOWS
CHATS	Every channel and every conversation with one node, newest first, with unread counts.
THREAD	The messages in one channel or conversation. Each of your messages shows its state: QUEUED, SENT, DELIVERED or FAILED, with the reason when it fails.
NODES	Every node the radio has heard. Tap one to start a direct conversation with it.
CONTACT	Give a node a nickname and a note. The nickname is shown everywhere instead of its short name.

6.2 USING IT

- Type in the box at the bottom and press Enter (or SEND) to send.
- DELETE removes a whole conversation. It asks for a second press within five seconds.
- Delivery states are honest: DELIVERED means the mesh heard the message, which is not proof that everyone got it.
- A message that arrives while you are on another screen shows as a notice on the hint line and as an unread count on the tile.

Messages are kept in MESHDECK/msg on the card, so they survive restarts. Your channels are the ones set in the radio.

7. MAP: OFFLINE MAPS AND PLACES

MAP draws offline Ordnance Survey maps from map packs on the card (MESHDECK/maps). It works with no internet.

7.1 THE SCREENS

SCREEN	WHAT IT DOES
MAP	The map with grid lines. Drag, or use the arrow keys, to move. The + and - buttons, or the + and - keys, zoom. C centres the map on you. S opens the search. Nodes show with their nicknames; you show as a filled dot.
SEARCH	Type a place name. The search covers the towns, villages, hamlets, hills and water in the place list you built for your area (section 7.2). A place found is marked on the map.
POSITION	Set where you are: type an OS grid reference (for example SX 923 928) or latitude and longitude, use the map centre, or use a node's GPS position.
NODES	Every node that has sent a position, with its distance and bearing from you.

> Map packs and the place list are not supplied in the download. You make them for your own area, as below.

7.2 GETTING MAPS FOR YOUR OWN AREA

You make maps on a PC from free Ordnance Survey data, with the Python tools in the repository's tools folder. The full guide, with every command, is docs/maps.md in the repository. The tools cover Great Britain only: there is nothing yet for Northern Ireland, the Isle of Man or the Channel Islands. The example below is Exeter; use your own area.

1. Download the data. From the OS Data Hub (osdatahub.os.uk, open data downloads), get OS Open Zoomstack in the Vector Tiles format (one .mbtiles file for all of Great Britain, a few gigabytes). It is free under the Open Government Licence.
2. Install the tools. Install Python 3.10 or newer, then run: `python -m pip install mapbox-vector-tile Pillow pyserial`
3. Find your area. The tools use OS grid eastings and northings in metres. Central Exeter is E 292300, N 92900 (grid reference SX 923 928). To convert a latitude and longitude, see docs/maps.md. For an area, note the south-west and north-east corners: a 50 km box around Exeter is 267000 68000 317000 118000.

4. Build the packs. Run these from the repository's top folder:
   - Great Britain overview (about 10 MB): `python tools/build_map.py ZOOMSTACK.mbtiles uk.map uk`
   - Detail for a town (a 20 km square, about 150 MB): `python tools/build_map.py ZOOMSTACK.mbtiles exeter.map local --centre 292300 92900 --size-km 20`
   - Or a larger area such as a county: `python tools/build_map.py ZOOMSTACK.mbtiles devon.map region --bbox 267000 68000 317000 118000`
5. Build the place list. `python tools/build_gazetteer.py ZOOMSTACK.mbtiles places.txt --box 267000 68000 317000 118000`. Always give the box: without it the tool uses a built-in default that is probably not your area.
6. Put them on the card. Copy the .map files and places.txt into MESHDECK/maps on the SD card (FAT32). Or, with the Tab5 connected by USB, send each file with `python tools/upload.py exeter.map exeter.map COM4` (use your own port). Restart the Tab5 if it was running.
7. Check. Open MAP (key 2): the map draws with the OS attribution at the bottom. Press S, type part of a town's name in your area, and choose it: the map centres on it. Zoom in with + to check the detail.

8. LIBRARY: REFERENCE CARDS AND BOOKS

8.1 CARDS

CARDS holds built-in UK reference cards: emergency contacts, three-day supplies, flood safety, power and gas safety, water interruption and food in a power cut. The cards are searchable. The contact numbers are reference information and are not a guarantee that a service is available.

8.2 BOOKS

BOOKS is the shelf of every book in MESHDECK/books. Books come in two kinds:

- PDF books are shown one page at a time, drawn in the theme colour.
- EPUB books are shown as text, one section at a time.

In the reader you can turn pages, open the contents, search inside the book, and go to a page number. MESHDECK remembers where you left each book. An EPUB can be made into a book on the deck: see section 10.

! Making PDF books. PDF books are made with a PC tool, tools/build_books.py, which is named in the library's empty-shelf message. That tool is not in the repository yet, so PDF books have to be made another way for now.

9. LOGBOOK: FIELD LOG, REPORTS, INVENTORY AND NOTES

LOGBOOK (tile 4) has four parts, opened by 1 to 4 or by tapping them.

9.1 LOG ENTRIES: THE FIELD LOG

- Type what happened, what was seen and where, then press ADD ENTRY. Each entry is stamped with the date and time (UK time, when the clock is set).
- Entries are added and never edited, so the record stays as it was written. The newest 200 are listed; all are kept.
- Entries are kept in MESHDECK/log/field.jnl. The file is designed so a power cut loses at most the entry being written.

9.2 REPORTS: STANDARD FORMS

Four forms follow the usual field mnemonics:

FORM	USE	BOXES
SITREP	Situation report	Who, where (grid reference, filled from your position when set), status, needs, next check-in, notes
METHANE	Major incident (JESIP)	Major incident declared, exact location, type, hazards, access, number of casualties, emergency services
ATMIST	Casualty handover	Age and sex, time of incident (filled from the clock), mechanism, injuries, signs, treatment
SALUTE	Observation	Size, activity, location (filled from your position), unit or identity, time seen (filled), equipment

Pick a form, tap a box and type. The preview shows the report as it will be saved. SAVE writes the report to MESHDECK/reports as a text file. Reports are not sent over the mesh; pass a saved report on by hand if you need to.

9.3 INVENTORY: KIT AND SUPPLY LISTS

- Keep a list per location (for example GRAB BAG, CAR, STORE CUPBOARD). Add a location with the box on the right.
- Add items with a count. Use -1 and +1 to change the count. DELETE asks for a second press.

- Give an item an expiry date as YYYY-MM-DD with SET DATE. Items expiring within 30 days, and expired items, are marked.
- STARTER KIT adds the fourteen items of the standard household emergency kit at zero, to be counted.
- CHECKED NOW records when a location was last checked. It refuses if the clock is not set.

The inventory is saved to MESHDECK/inventory.

9.4 NOTES

Plain-text notes, one file each, in MESHDECK/notes. Type a note and press SAVE NOTE. Tap a note to read it. DELETE removes a note after a second press. Notes are text only; use the reports for forms.

10. FILES: YOUR FILES, SCREENSHOTS AND IMPORTS

FILES browses MESHDECK/Files on the card. The Inbox and Screenshots folders are at the top.

ACTION	HOW
Open a folder or select a file	Enter, or tap.
See the details	The name, size and a text preview of the first 6 KB. Pictures show a note instead.
Rename	Type the new name and press RENAME.
Copy	COPY makes a copy beside the file, named with -copy.
Delete	Press DELETE, then CONFIRM DELETE within five seconds. Folders are not deleted here.
View a picture	PNG, JPEG and BMP files show a VIEW IMAGE button. Esc closes the picture.
Import an EPUB as a book	Select an EPUB (for example one copied into the Inbox) and press IMPORT AS BOOK. The book is saved to MESHDECK/books and appears on the LIBRARY shelf.

The import reads the whole book before it writes, so a long book takes a few seconds and the screen waits. Very long books are cut short at about 4 MB of text, and the screen says so. Only EPUB files are imported; PDF books are not imported on the deck yet.

11. TERMINAL: TYPED COMMANDS

TERMINAL is a command line. Type a command and press Enter or RUN. The answer appears above the command line, and the newest lines are kept. Command names are not case-sensitive.

COMMAND	WHAT IT REPORTS
HELP	The list of commands, with what each does.
STATUS	The version, the device, the time since start-up, free memory and whether the card is present.
POWER	Battery percentage, voltage, average current, external power and the time left. Values not measured are said to be not measured.
MESH	The radio link, the channels, the number of nodes heard and the messages kept.
NODES	The nodes heard most recently, with signal (SNR) and how long ago.
TIME	The clock in UK time, or CLOCK NOT SET.
SCREENSHOT	Saves the screen to MESHDECK/Files/Screenshots.
CLEAR	Empties the terminal screen.

An unknown word gets "UNKNOWN COMMAND ... TYPE HELP". The terminal only reports; it changes no settings and no data.

12. SETTINGS: DISPLAY, CLOCK AND ABOUT

PART	WHAT IT DOES
1 DISPLAY	The colour theme (green, amber or red), the brightness from 10 to 100 percent with + and -, and TAKE SCREENSHOT.
2 CLOCK	Set the date and time. UK local time, with GMT or BST worked out automatically.
3 ABOUT	The version, the platform, the storage in use, and the licences. CREDITS opens the author's logo and a short note.

Brightness is the only power setting. There is no sleep mode, no screen-off and no power-saving preset (section 16).

13. DIAGNOSTICS: POWER, SYSTEM AND THE TEST PAGES

DIAGNOSTICS shows the live power figures: battery voltage and current, the one-minute average, and the estimated time left. It also shows the system basics: the version, the storage and the memory.

Three buttons along the bottom open the test pages. TOUCH TEST shows a grid of boxes; tap each one and it turns solid, so a dead spot on the screen stays empty. KEY LOG lists the last keys pressed, newest first, with their codes. RADIO shows the mesh link, your node number, how many nodes have been heard, and the channels. BACK or Esc returns to the main page.

14. GAMES: RALLY, SEA BATTLE AND KLONDIKE

GAMES (tile 0) is a hub of three games with your best score for each. Press 1, 2 or 3, or tap a game. Best scores are kept on the card in MESHDECK/games/scores.txt, and only a new best is written.

14.1 RALLY: PONG AGAINST THE CPU

- You play the left paddle against the CPU on the right. First to seven points wins the match.
- Keys: W and S, or the up and down arrows, move the paddle. A key press keeps it moving briefly, since the keyboard sends presses, not holds.
- SPACE or Enter starts a match, and starts a new one after a match ends.
- The ball speeds up a little with each return. The CPU is a little slower than a quick hand and aims slightly off, so it can be beaten.
- Best score: the most points you scored in one match.

14.2 SEA BATTLE: FLEETS AGAINST THE CPU

- Each side has five ships (lengths 5, 4, 3, 3 and 2) on a 10 by 10 sea. Both fleets are placed at random when a game starts.
- The left sea is YOUR FLEET, where the CPU fires. The right sea is the CPU's sea, where you fire.
- Move the cursor with the arrows and press Enter or Space to fire. You can also tap a square. The CPU fires back after each of your shots, and once it hits, it fires next to its hits until the ship sinks.
- N starts a new fleet. A win is scored as 200 minus the number of shots, so a higher score is better. Only won games are saved.

Fleets are placed at random. You cannot place your own ships yet.

14.3 KLONDIKE: DRAW ONE

- The standard solitaire layout: a stock and a waste pile at the top left, four foundations at the top right, and seven columns below.

- D, or Enter on the stock, turns one card from the stock onto the waste. When the stock is empty, the waste turns back over into the stock.
- Move the cursor over the 13 places with the arrows. Enter picks up a card (or a run of face-up cards, from a column) and Enter on another place puts it down if the rules allow. Up and down choose how much of a run to take.
- Cards build down in alternating colours in the columns, and up by suit from the ace in the foundations. Only kings start an empty column.
- U undoes the last move or draw. N deals a new game.
- Best score: the fewest moves to win, kept as 1000 minus the moves. Only a won game is saved.

KLONDIKE is played with the keyboard only at present. Tapping cards is not built yet.

14A. TOOLS: TIMER, STOPWATCH AND CALCULATOR

TOOLS (tile A) is a hub of three small tools. Press 1, 2 or 3, or tap one. Nothing here is saved to the card.

14A.1 TIMER

- Pick 1, 5, 10 or 15 minutes, then START. SPACE starts or pauses. R resets to the chosen length.
- The status line shows TIME'S UP when the time runs out.

14A.2 STOPWATCH

- START and STOP (or SPACE). LAP records the time so far, and the last eight laps are listed. RESET clears everything.
- Tenths of a second show while it runs.

14A.3 CALCULATOR

- Digits, the point, and + - * / , then = or Enter to work it out. It works left to right, as a pocket calculator does: 2 + 3 * 4 gives 20.
- BKSP deletes the last digit, and C clears everything. Division by zero shows ERROR until you press C.
- Up to 12 digits can be typed. There is no key for a minus sign on a number you type, but a result can be negative (1 - 5 gives -4).

15. THE BOOT SPLASH AND THE HOME INFORMATION PANEL

15.1 THE BOOT SPLASH

At every power-on, green drips fall down the screen and light up the words MESHDECK - OS, letter by letter. When the words are complete, the drips still falling run off the bottom of the screen, leaving the words alone on black; then the words fade out, and the lock screen (if a PIN is set), the setup (on first start) or the home screen appears. Any key or a tap skips the splash.

15.2 THE HOME INFORMATION PANEL

The empty space beside the tiles shows, live:

- The mesh state: MESH OK, MESH CONNECTING, or NO RADIO.
- How many nodes were heard in the last two hours, and the two strongest signals by SNR (signal-to-noise ratio, in dB).
- The battery percentage, free memory, uptime and the number of mesh channels.

16. POWER: WHAT THE DECK DOES AND DOES NOT DO

MESHDECK's one power control is the screen brightness, set in SETTINGS › DISPLAY, from 10 to 100 percent.

> The deck does not sleep, does not switch off the screen, does not slow the processor, does not switch parts off, and has no power presets. That was a deliberate decision. The screen stays on, and nothing is powered down without you asking.

DIAGNOSTICS and the POWER command give the battery figures. The battery time left is an estimate from the average current over the last minute, and is shown as unknown until enough readings exist.

17. THE SD CARD: WHERE EVERYTHING IS KEPT

Everything you save is on the card, under the MESHDECK folder:

FOLDER	HOLDS
MESHDECK/ (root)	Settings and the position you set.
MESHDECK/log/	The system journal and the field log (field.jnl).
MESHDECK/msg/	Messages and contacts (nicknames and notes).
MESHDECK/maps/	Map packs (*.map) and the place list.
MESHDECK/books/	Book packs (*.bk) and where you left each book.
MESHDECK/reports/	Saved reports.
MESHDECK/inventory/	The inventory lists.
MESHDECK/notes/	Plain-text notes.
MESHDECK/games/	Best scores.
MESHDECK/Files/	Your own files, visible in FILES, with Inbox and Screenshots folders.
MESHDECK/update/	Reserved for signed firmware updates.

The PIN is not on the card. It is stored in the Tab5's internal flash, so removing the card does not change it. The card is not encrypted (section 5). Files are written so that a power cut leaves either the old file or the new one.

18. PRIVACY: WHAT LEAVES THE DECK, AND WHAT DOESN'T

- Nothing leaves the deck by itself. MESHDECK has no internet connection and sends no data to any service.
- Mesh messages leave the deck only when you send them. Other nodes can read a channel message, so use direct messages for anything private, and remember that anything sent may be kept by the people who receive it.
- Your position is saved on the card when you set it. It is not sent unless you put it in a report and send that, or share it on the mesh yourself. MESHDECK does not broadcast your position.
- Positions of other nodes are received and shown on the map, but not passed on.
- Map and place data on the card is only what you copied there.

! Before sharing a screenshot, a report, a notes file or a card, check it for places, grid references and names. The screenshots include whatever is on screen, and Ctrl+P does not blur anything.

19. TROUBLESHOOTING

WHAT YOU SEE	WHAT TO DO
NO RADIO in MESSAGES or the status bar	Plug the Unit C6L into the USB-A port and wait a few seconds. MESH CONNECTING means it is starting.
NO STORAGE, or an empty FILES, MAP or LIBRARY	The card is missing or not readable. Check it is FAT32 and fully inserted, then restart.
NO KEYBOARD in the status bar	The physical keyboard is not connected. Touch still works for tiles and buttons.
LOCKED	Type the PIN. After several wrong PINs, wait for the time shown before trying again.
A screen does not respond to keys	Press Esc to go back one step, then try again. If the whole deck stops responding, restart it with the Tab5's power switch or reset button.
NOT IMPORTED: ... when importing an EPUB	The message gives the reason (for example, no container file, or no readable text). The file is not changed.
NO MAP PACK, or NO PLACE INDEX	No .map file, or no places.txt, in MESHDECK/maps. Check the folder and file names (section 7.2), then restart.
NO MAP DATA HERE AT THIS ZOOM	The view is outside your map packs. Zoom out, or search for a place in your area.
The map is slow the first time an area is shown	The first view reads the tiles from the card. Later views of the same area are quick.

20. CAPABILITIES AT A GLANCE

AREA	CAPABILITY
Shell	Eleven-tile home, key and touch navigation, status bar, hint line, screenshots, colour themes, boot splash, home information panel.
Security	PIN (4 to 12 digits), salted hash, auto-lock, lock now, wrong-PIN waits.
Messages	Channel and direct messages over Meshtastic, delivery states, nodes, nicknames, delete conversations.
Map	Offline OS maps with zoom and pan, grid lines, place search, setting your position, node positions and distances.
Library	Built-in reference cards, PDF and EPUB book reader with contents, search and bookmarks.
Logbook	Field log, four report forms saved on the card, inventory with expiry, notes.
Files	Browse, rename, copy, delete, view pictures, EPUB import as a book.
Terminal	Eight commands with help text.
Settings and diagnostics	Theme, brightness, clock, about and licences; live battery and system figures; touch test, key log and radio pages.
Tools	Timer, stopwatch and calculator.
Games	RALLY, SEA BATTLE and KLONDIKE, with best scores on the card.

21. FOR DEVELOPERS: BUILDING, TESTING AND THE CODE LAYOUT

21.1 BUILDING

MESHDECK's core is plain C++17 with no Arduino or ESP-IDF code, shared by the Tab5 firmware, a desktop simulator and the unit tests. The repository's README gives the commands:

- `cmake -S . -B build` then `cmake --build build`, and `ctest --test-dir build` for the tests.
- The simulator is `build/platform/sim/meshdeck-sim`. It runs the same screens in a window. The Tab5 firmware builds with PlatformIO: `pio run -d platform/tab5`.

21.2 CODE LAYOUT

FOLDER	CONTENTS
core/kernel/	Events, settings, storage and paths, the journal (append-only log), the clock.
core/services/	Mesh, position, power, security, contacts, inventory, terminal, high scores.
core/formats/	Map packs, book packs, images, the zip reader, EPUB import, reports, OS grid conversion.
core/games/	The rules of RALLY (pong), SEA BATTLE and KLONDIKE, with no screen code.
core/ui/ and core/apps/	The shell, the overlays (lock, setup, splash) and each app's screens.
platform/tab5/	The Tab5 firmware: display, keyboard, touch, radio, battery, SD card, the EPUB and PPA drivers.
platform/sim/	The desktop simulator.
tests/	Unit tests, one file per area (more than 100 cases).
docs/	Architecture, file formats (journal, map pack, book pack), hardware notes, power, security, roadmap and this manual.

21.3 STANDARDS THE CODE FOLLOWS

- Every value that is not known is shown as not known. No invented numbers.
- Writes to the card are safe against power loss: a temporary file is written and then renamed.
- Game and calculation rules are kept apart from the screens, so they can be tested without a display.
- MESHDECK's own code builds with no compiler warnings. A full clean build of the Tab5 firmware can still show one warning from the vendor SPI driver.

22. LICENCE, CREDITS AND THIRD-PARTY SOFTWARE

MESHDECK is released under the MIT licence (see the LICENSE file). Bundled libraries keep their own licences, listed in THIRD_PARTY.md. The embedded fonts and graphics are used under their own terms.

Map data: contains OS data © Crown copyright and database right, used under the Open Government Licence.

MESHDECK was created by RabzLabz and Claude. It is a civilian resilience application, aimed to help civilians during times of natural disaster and civil unrest. Stay prepared, stay strong.